

4.单目运算符重载

日期相减

C++

```
1  #include <iostream>
2  // 定义日期类
3  class date {
4  private:
5      int y; // 年份
6      int m; // 月份
7      int d; // 日份
8  public:
9      // 构造函数，初始化年、月、日，默认为 0
10     date(int year = 0, int month = 0, int day = 0) : y(year), m(month), d(day) {}
11     // 返回年份
12     int year() const {
13         return y;
14     }
15     // 返回月份
16     int month() const {
17         return m;
18     }
19     // 返回日份
20     int day() const {
21         return d;
22     }
23     // 重载减法运算符，计算两个日期之间的天数差
24     int operator-(const date& other) const {
25         int days = (y - other.y) * 365 + (m - other.m) * 30 + (d - other.d);
26         return days;
27     }
28     // 重载复合赋值运算符 "+=", 将日期增加指定天数
29     date& operator+=(int days) {
30         d += days;
31         while (d > 30) { // 如果日份大于 30，说明需要进位到下个月
32             d -= 30;
33             m++; // 月份加一
34             if (m > 12) { // 如果月份大于 12，说明需要进位到下一年
35                 m = 1; // 月份重置为 1
36                 y++; // 年份加一
37             }
38         }
39         return *this; // 返回当前对象的引用
40     }
41     // 重载复合赋值运算符 "-=", 将日期减少指定天数
42     date& operator-=(int days) {
43         d -= days;
44         while (d < 1) { // 如果日份小于 1，说明需要借位到上个月
45             d += 30;
46             m--; // 月份减一
47             if (m < 1) { // 如果月份小于 1，说明需要借位到上一年
48                 m = 12; // 月份重置为 12
49                 y--; // 年份减一
50             }
51         }
52         return *this; // 返回当前对象的引用
53     }
54 };
55 // 主函数
56 int main(void) {
57     using std::cin;
58     using std::cout;
```

```

59     using std::endl;
60     date D1, D2;    // 定义两个日期对象
61     int year, month, day;
62     cin >> year >> month >> day;    // 输入第一个日期
63     D1 = date(year, month, day);    // 创建第一个日期对象
64     cin >> year >> month >> day;    // 输入第二个日期
65     D2 = date(year, month, day);    // 创建第二个日期对象
66
67     cout << D2 - D1 << endl;    // 输出两个日期之间的天数差
68
69     cin >> day;    // 输入要增加或减少的天数
70     D1 += day;    // 将第一个日期增加指定天数
71     D2 -= day;    // 将第二个日期减少指定天数
72     // 输出两个日期增加或减少天数后的年、月、日
73     cout << D1.year() << " " << D1.month() << " " << D1.day() << endl;
74     cout << D2.year() << " " << D2.month() << " " << D2.day() << endl;
75
76     return 0;
77 }

```

日期相减 2

C++

```

1  #include <iostream>
2
3  // 定义日期类
4  class date {
5  private:
6      int y;    // 年份
7      int m;    // 月份
8      int d;    // 日份
9  public:
10     // 构造函数，初始化年、月、日，默认为 0
11     date(int year = 0, int month = 0, int day = 0) : y(year), m(month), d(day) {}
12
13     // 返回年份
14     int year() const {
15         return y;
16     }
17
18     // 返回月份
19     int month() const {
20         return m;
21     }
22
23     // 返回日份
24     int day() const {
25         return d;
26     }
27
28     // 判断是否是闰年
29     bool isLeapYear(int year) const {
30         return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
31     }
32
33     // 获取指定年月的天数
34     int daysInMonth(int year, int month) const {
35         if (month == 2 && isLeapYear(year)) {
36             return 29;
37         } else {
38             int days[13] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
39             return days[month];
40         }
41     }
42
43     // 重载减法运算符，计算两个日期之间的天数差
44     int operator-(const date& other) const {
45         int days = 0;
46         // 计算两个日期之间的年份差
47         for (int i = other.y; i < y; ++i) {

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48         days += isLeapYear(i) ? 366 : 365;
49     }
50     // 计算两个日期之间的月份差
51     for (int i = 1; i < m; ++i) {
52         days += daysInMonth(y, i);
53     }
54     days += d;
55     // 减去被减日期的天数
56     for (int i = 1; i < other.m; ++i) {
57         days -= daysInMonth(other.y, i);
58     }
59     days -= other.d;
60     return days;
61 }
62
63 // 重载复合赋值运算符 "+=", 将日期增加指定天数
64 date& operator+=(int days) {
65     while (days > 0) {
66         int daysInCurrentMonth = daysInMonth(y, m);
67         if (d + days > daysInCurrentMonth) {
68             days -= daysInCurrentMonth - d + 1;
69             d = 1;
70             if (m == 12) {
71                 m = 1;
72                 ++y;
73             } else {
74                 ++m;
75             }
76         } else {
77             d += days;
78             days = 0;
79         }
80     }
81     return *this;
82 }
83
84 // 重载复合赋值运算符 "-=", 将日期减少指定天数
85 date& operator-=(int days) {
86     while (days > 0) {
87         if (d - days < 1) {
88             days -= d;
89             if (m == 1) {
90                 m = 12;
91                 --y;
92             } else {
93                 --m;
94             }
95             d = daysInMonth(y, m);
96         } else {
97             d -= days;
98             days = 0;
99         }
100     }
101     return *this;
102 }
103 };
104 // 主函数
105 int main(void) {
106     using std::cin;
107     using std::cout;
108     using std::endl;
109
110     date D1, D2; // 定义两个日期对象
111     int year, month, day;
112
113     cin >> year >> month >> day; // 输入第一个日期
114     D1 = date(year, month, day); // 创建第一个日期对象
115     cin >> year >> month >> day; // 输入第二个日期
116     D2 = date(year, month, day); // 创建第二个日期对象
117

```

```

118     cout << D2 - D1 << endl;    // 输出两个日期之间的天数差
119
120     cin >> day;    // 输入要增加或减少的天数
121     D1 += day;    // 将第一个日期增加指定天数
122     D2 -= day;    // 将第二个日期减少指定天数
123
124     // 输出两个日期增加或减少天数后的年、月、日
125     cout << D1.year() << " " << D1.month() << " " << D1.day() << endl;
126     cout << D
127

```

时间上一秒

```

1  //StudybarCommentBegin
2  #include <iostream>
3  #include <iomanip>
4  using std::cin;
5  using std::cout;
6  using std::endl;
7  using std::setfill;
8  using std::setw;
9  //StudybarCommentEnd
10 class Time {
11 private:
12     int hour;
13     int minute;
14     int second;
15
16 public:
17     Time(int h = 0, int m = 0, int s = 0) : hour(h), minute(m), second(s) {}
18
19     void setTime(int h, int m, int s) {
20         hour = h;
21         minute = m;
22         second = s;
23     }
24     void printTime() const {
25         // 设置填充字符为 '0', 确保不足两位的数字用 '0' 填充
26         std::cout << std::setfill('0');
27         // 输出小时部分, 确保输出长度为两位数
28         std::cout << std::setw(2) << hour << ":";
29         // 输出分钟部分, 确保输出长度为两位数
30         std::cout << std::setw(2) << minute << ":";
31         // 输出秒钟部分, 确保输出长度为两位数
32         std::cout << std::setw(2) << second << std::endl;
33     }
34     Time& operator--() {
35         if (second == 0) {
36             if (minute == 0) {
37                 hour = (hour - 1 + 24) % 24;
38                 minute = 59;
39             } else {
40                 --minute;
41             }
42             second = 59;
43         } else {
44             --second;
45         }
46         return *this;
47     }
48
49     // 后置递减运算符重载函数
50     Time operator--(int) {
51         Time temp = *this; // 复制当前对象
52         --(*this); // 调用前置递减运算符
53         return temp; // 返回递减前的对象
54     }
55
56     Time operator+(int seconds) const {

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C++

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57         Time result = *this;
58         if (seconds ≥ 0) {
59             result.second += seconds;
60
61             result.minute += result.second / 60;
62             result.second %= 60;
63
64             result.hour += result.minute / 60;
65             result.minute %= 60;
66
67             result.hour %= 24;
68         } else {
69             seconds = -seconds;
70             int totalSeconds = hour * 3600 + minute * 60 + second;
71             totalSeconds -= seconds;
72             if (totalSeconds < 0) {
73                 totalSeconds += 24 * 3600;
74             }
75             result.hour = totalSeconds / 3600;
76             totalSeconds %= 3600;
77             result.minute = totalSeconds / 60;
78             result.second = totalSeconds % 60;
79         }
80         return result;
81     }
82
83     friend Time operator+(int seconds, const Time& t);
84 };
85
86 Time operator+(int seconds, const Time& t) {
87     return t + seconds;
88 }
89 //StudybarCommentBegin
90 int main()
91 {
92     int hour, minute, second;
93     int number;
94     Time t1(23, 45, 0), t2, t3(t1), t4;
95     cin >> hour >> minute >> second >> number;
96     t1.setTime(hour, minute, second);
97     t2 = t1--;
98     t2.printTime();
99     t1.printTime();
100    t3 = --t1;
101    t3.printTime();
102    t4 = number + t1;
103    t4.printTime();
104 }
105 //StudybarCommentEnd

```

时间下一秒

```

1  //StudybarCommentBegin
2  #include <iostream>
3  #include <iomanip>
4  using std::cin;
5  using std::cout;
6  using std::endl;
7  using std::setfill;
8  using std::setw;
9  //StudybarCommentEnd
10 class Time {
11 private:
12     int hour;
13     int minute;
14     int second;
15
16 public:
17     Time(int h = 0, int m = 0, int s = 0) : hour(h), minute(m), second(s) {}

```

C++

```

18
19 // 设置时间
20 void setTime(int h, int m, int s) {
21     hour = h;
22     minute = m;
23     second = s;
24 }
25
26 // 打印时间
27 void printTime() const {
28     std::cout << setfill('0') << setw(2) << hour << ":"
29         << setw(2) << minute << ":"
30         << setw(2) << second << std::endl;
31 }
32
33 // 重载加法操作符
34 Time operator+(int seconds) const {
35     Time result = *this;
36     result.second += seconds;
37
38     if (result.second ≥ 60) {
39         result.minute += result.second / 60;
40         result.second %= 60;
41
42         if (result.minute ≥ 60) {
43             result.hour += result.minute / 60;
44             result.minute %= 60;
45
46             result.hour %= 24;
47         }
48     }
49
50     return result;
51 }
52
53 // 前置++操作符
54 Time& operator++() {
55     *this = *this + 1;
56     return *this;
57 }
58 friend Time operator+(int seconds, const Time& t);
59 };
60
61 // 友元函数的实现
62 Time operator+(int seconds, const Time& t) {
63     Time result = t;
64     result.second += seconds;
65
66     if (result.second ≥ 60) {
67         result.minute += result.second / 60;
68         result.second %= 60;
69
70         if (result.minute ≥ 60) {
71             result.hour += result.minute / 60;
72             result.minute %= 60;
73
74             result.hour %= 24;
75         }
76     }
77     return result;
78
79 //StudybarCommentBegin
80 int main()
81 {
82     int hour, minute, second;
83     int increase;
84     Time t1(23, 45, 0), t2, t3(t1);
85     cin >> hour >> minute >> second >> increase;
86     t1.setTime(hour, minute, second);
87     t1.printTime();

```

```

88     t2=++t1;
89     t2.printTime();
90     t3=increase+t1;
91     t3.printTime();
92     return 0;
93 }
94 //StudybarCommentEnd

```

时间下一秒 ++

C++

```

1  //StudybarCommentBegin
2  #include <iostream>
3  #include <iomanip>
4  using std::cin;
5  using std::cout;
6  using std::endl;
7  using std::setfill;
8  using std::setw;
9  //StudybarCommentEnd
10 class Time {
11 private:
12     int hour;
13     int minute;
14     int second;
15
16 public:
17     Time(int h = 0, int m = 0, int s = 0) : hour(h), minute(m), second(s) {}
18
19     void setTime(int h, int m, int s) {
20         hour = h;
21         minute = m;
22         second = s;
23     }
24
25     void printTime() const {
26         std::cout << std::setfill('0') << std::setw(2) << hour << ":"
27             << std::setw(2) << minute << ":"
28             << std::setw(2) << second << std::endl;
29     }
30
31     Time& operator--() {
32         if (second == 0) {
33             if (minute == 0) {
34                 hour = (hour - 1 + 24) % 24;
35                 minute = 59;
36             } else {
37                 --minute;
38             }
39             second = 59;
40         } else {
41             --second;
42         }
43         return *this;
44     }
45
46     Time operator--(int) {
47         Time temp = *this;
48         --(*this);
49         return temp;
50     }
51     Time& operator++() {
52         *this = *this + 1;
53         return *this;
54     }
55
56     Time operator++(int) {
57         Time temp = *this;
58         ++(*this);
59         return temp;

```

```

60     }
61     Time operator+(int seconds) const {
62         Time result = *this;
63         if (seconds >= 0) {
64             result.second += seconds;
65
66             result.minute += result.second / 60;
67             result.second %= 60;
68
69             result.hour += result.minute / 60;
70             result.minute %= 60;
71
72             result.hour %= 24;
73         } else {
74             seconds = -seconds;
75             int totalSeconds = hour * 3600 + minute * 60 + second;
76             totalSeconds -= seconds;
77             if (totalSeconds < 0) {
78                 totalSeconds += 24 * 3600;
79             }
80             result.hour = totalSeconds / 3600;
81             totalSeconds %= 3600;
82             result.minute = totalSeconds / 60;
83             result.second = totalSeconds % 60;
84         }
85         return result;}
86     friend Time operator+(int seconds, const Time& t) {
87         return t + seconds;}
88 };
89
90
91 //StudybarCommentBegin
92 int main()
93 {
94     int hour, minute, second;
95     int increase;
96     Time t1(23, 45, 0), t2, t3(t1);
97     cin >> hour >> minute >> second >> increase;
98     t1.setTime(hour, minute, second);
99     t1.printTime();
100    t2 = ++t1;        //This is for ++t1
101    t2.printTime();
102
103    t3 = increase + t1; //This is for friend function
104    t3.printTime();
105
106    t1 = t2++;        //This is for t2++
107    t1.printTime();
108    t2.printTime();
109
110    t1 = (++t3)++;    // This is for left value (return the reference of the object )
111    t1.printTime();
112    t3.printTime();
113    return 0;
114 }
115 //StudybarCommentEnd

```

下一天

```

1  #include <iostream>
2
3  class date {
4  private:
5      int y;
6      int m;
7      int d;
8
9  public:
10     date(int year = 0, int month = 0, int day = 0) : y(year), m(month), d(day) {}

```

C++

```

11
12     int year() const {
13         return y;
14     }
15
16     int month() const {
17         return m;
18     }
19
20     int day() const {
21         return d;
22     }
23     date operator++(int) {
24         date temp = *this;
25         if (d == 31 && (m == 1 || m == 3 || m == 5 || m == 7 || m == 8 || m == 10 || m == 12)) {
26             if (m == 12) {
27                 y++;
28                 m = 1;
29             } else {
30                 m++;
31             }
32             d = 1;
33         } else if (d == 30 && (m == 4 || m == 6 || m == 9 || m == 11)) {
34             m++;
35             d = 1;
36         } else if (d == 28 && m == 2 && !isLeapYear()) {
37             m++;
38             d = 1;
39         } else if (d == 29 && m == 2 && isLeapYear()) {
40             m++;
41             d = 1;
42         } else {
43             d++;
44         }
45         return temp;
46     }
47
48     bool isLeapYear() const {
49         return ((y % 4 == 0 && y % 100 != 0) || (y % 400 == 0));
50     }
51 };
52
53 //StudybarCommentBegin
54 int main(void)
55 {
56     using std::cin;
57     using std::cout;
58     using std::endl;
59
60     date D1,D2;
61
62     int year;
63     int month;
64     int day;
65
66     cin>>year>>month>>day; //shu ru di yi ge ri qi
67     D1=date(year,month,day);
68
69     D2=D1++;
70
71     cout<<D1.year()<<" "<<D1.month()<<" "<<D1.day()<<endl;
72     cout<<D2.year()<<" "<<D2.month()<<" "<<D2.day()<<endl;
73     return 0;
74 }
75 //StudybarCommentEnd

```